ENME601 Thermodynamics

Week 3 – Properties of pure substances

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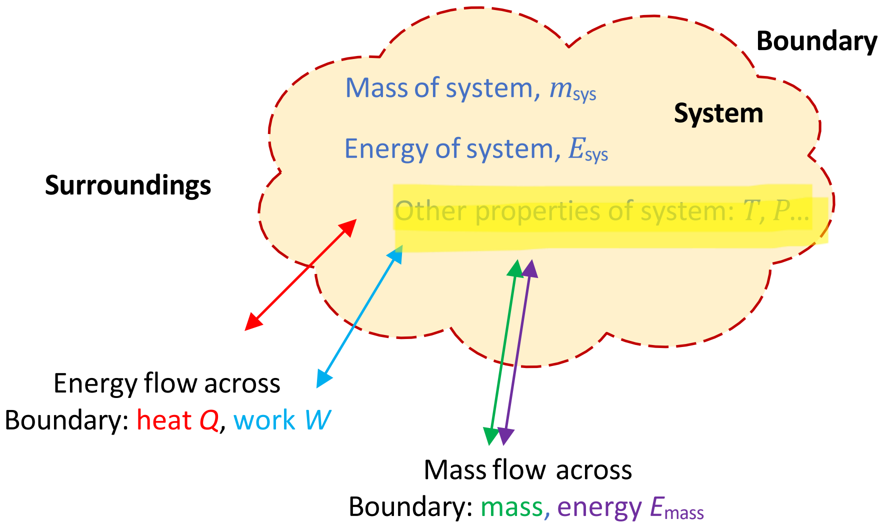
Introduction

- We have seen that energy interactions across the boundary cause energy changes within the system and that energy must be conserved (first law).
- However, these energy and mass interactions also cause *other properties* (apart from energy) of the substance or material in the system *to change*.
- We revise the concept of a *pure substance* and discuss the physics of *phase change processes* of pure substances.
- We also consider the *ideal gas equation* which relates temperature, pressure and volume for ideal gases

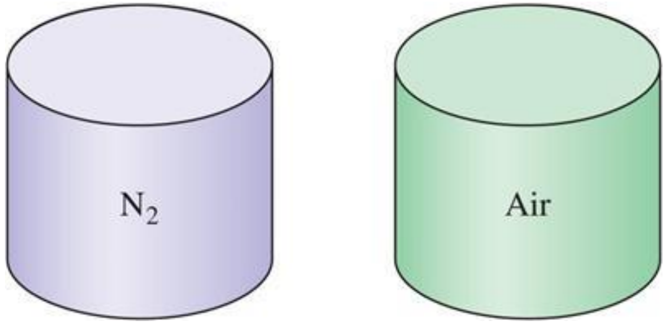


Pure Substance

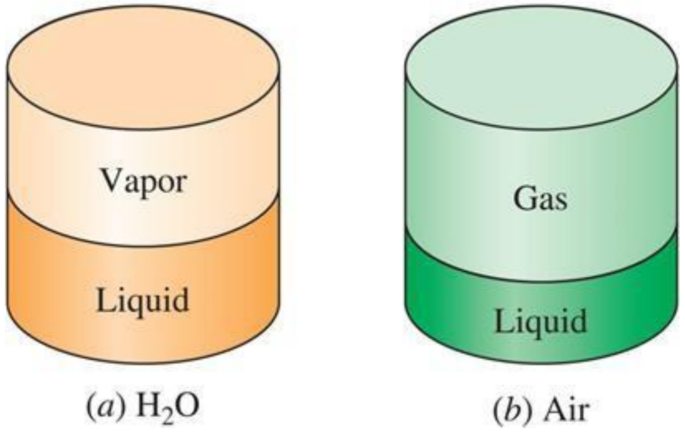
- **Pure substance:** A substance that has a fixed chemical composition throughout.

  Examples

- a single chemical element: Nitrogen, copper
- a compound: water, carbon dioxide
- a mixture of elements and compounds: air
- mixture of phases: ice and liquid water (as long as chemical composition of all the phases is identical)



Nitrogen and gaseous air are pure substances



A mixture of liquid and gaseous water is a pure substance, but a mixture of liquid and gaseous air is not

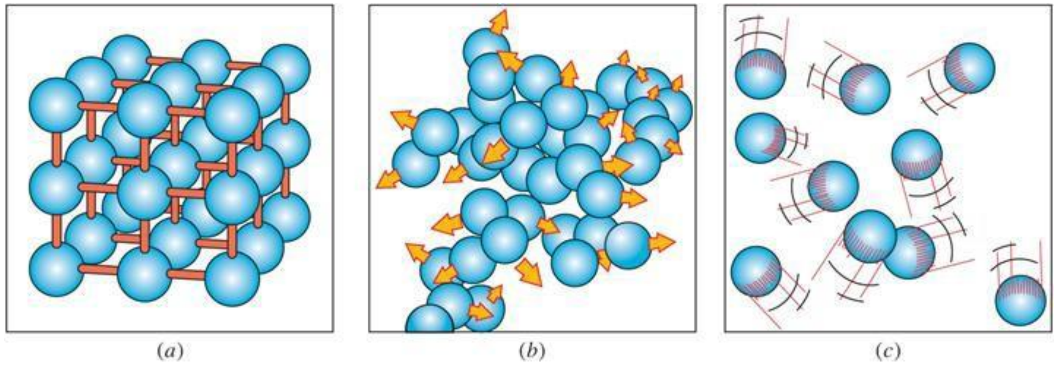
Phases of a Pure Substance

- A **phase of a substance** is identified as having a *distinct molecular arrangement* that is homogenous throughout.
- The three principal phases: **solid, liquid and gas**
- A pure substance may have different phases within a principal phase e.g. iron has three solid phases (bcc, fcc, hcp), carbon has two solid phases (graphite, diamond)
- The arrangement of atoms in different phases:

  (a) Solid: molecules are at relatively fixed positions

  (b) Liquid: groups of molecules move about each other

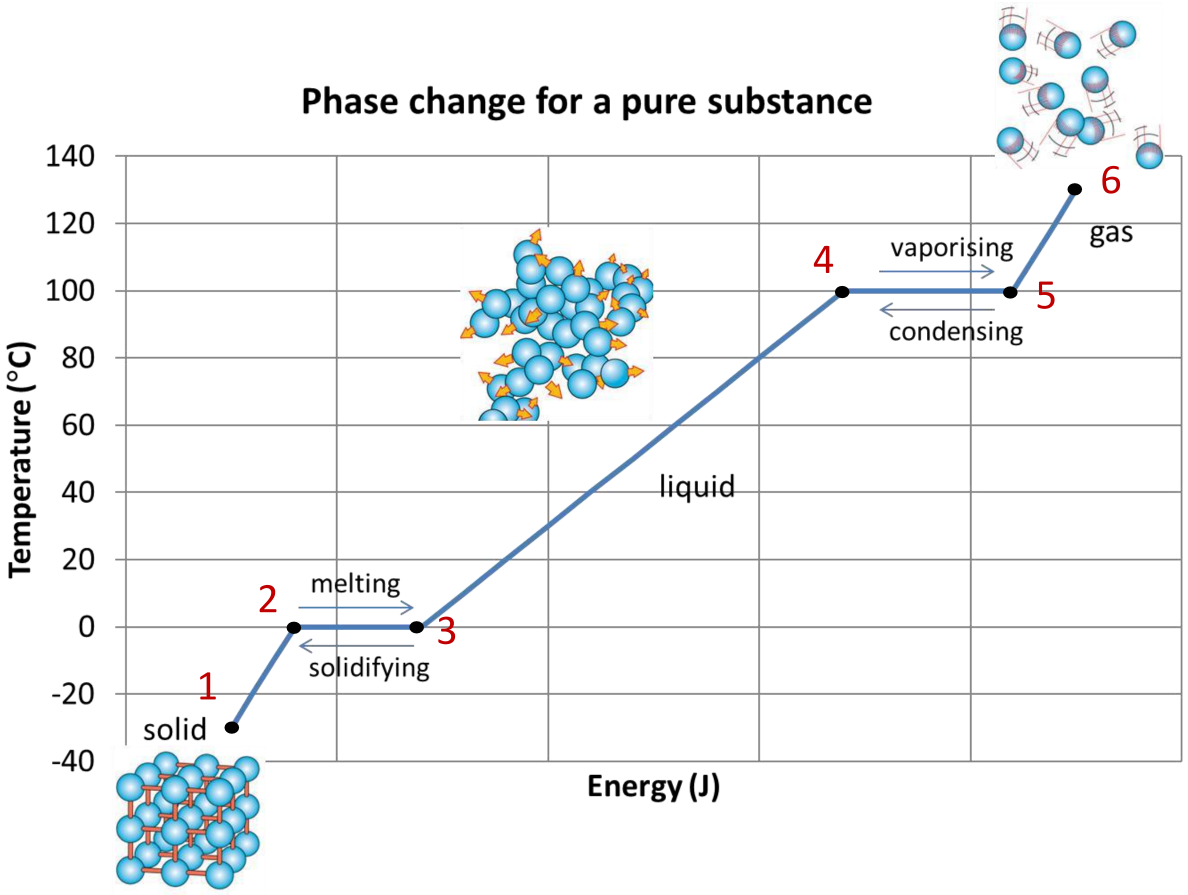
  (c) Gas: molecules move about at random



Solid　　Liquid　　Gas

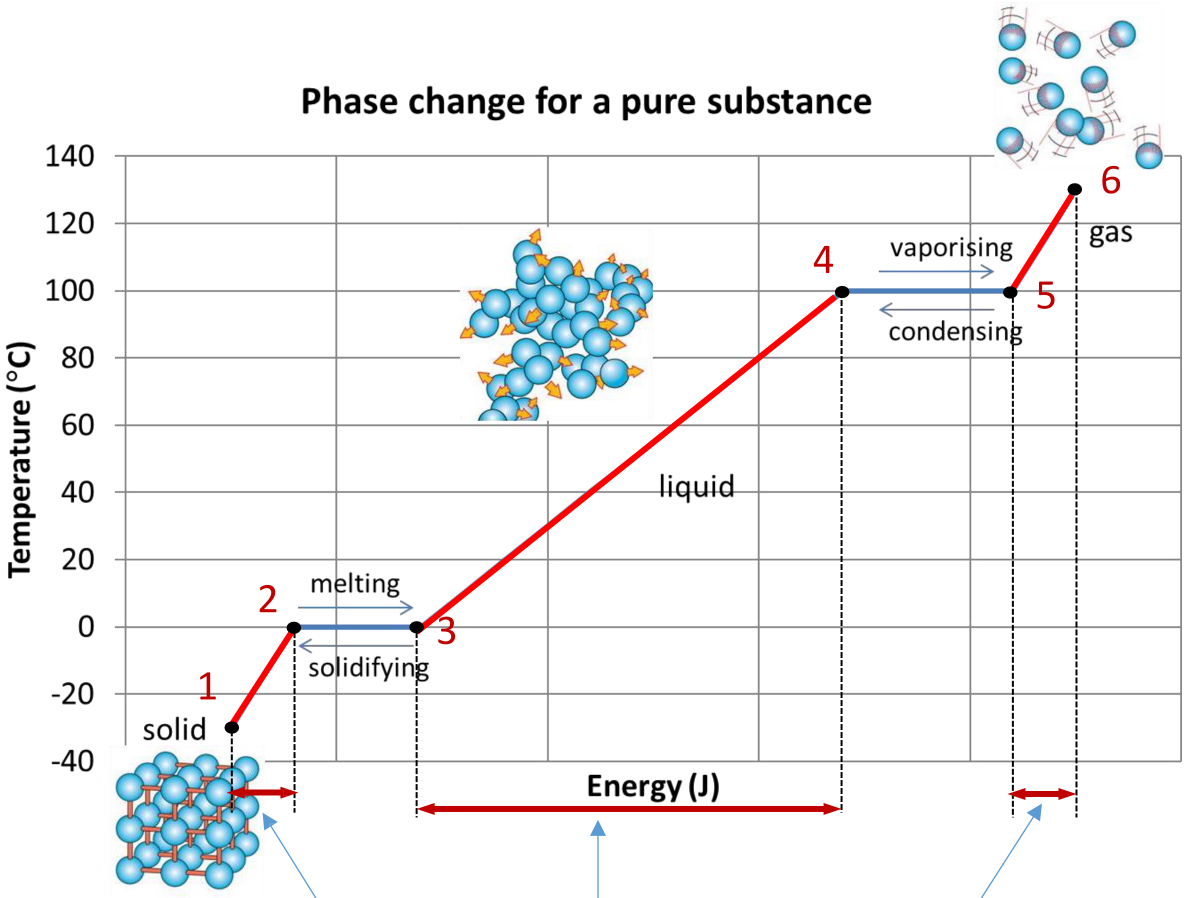
Phase-Change Processes of Pure Substances

- Consider a pure substance initially as a solid (1).
- As **energy is added** to the solid the **internal energy of the substance increases. (process from 1→6)**
  - 1→2 - The solid increases in temperature till it begins melting
  - 2→3 - The solid melts till it is all liquid
  - 3→4 - The liquid increases in temperature till it begins to vaporise
  - 4→5 - The liquid vaporises till it is all gas
  - 5→6 - The gas heats up…
- If energy is removed from the hot gas (at 6) the internal energy decreases and the process is reversed



Phase-Change Processes of Pure Substances

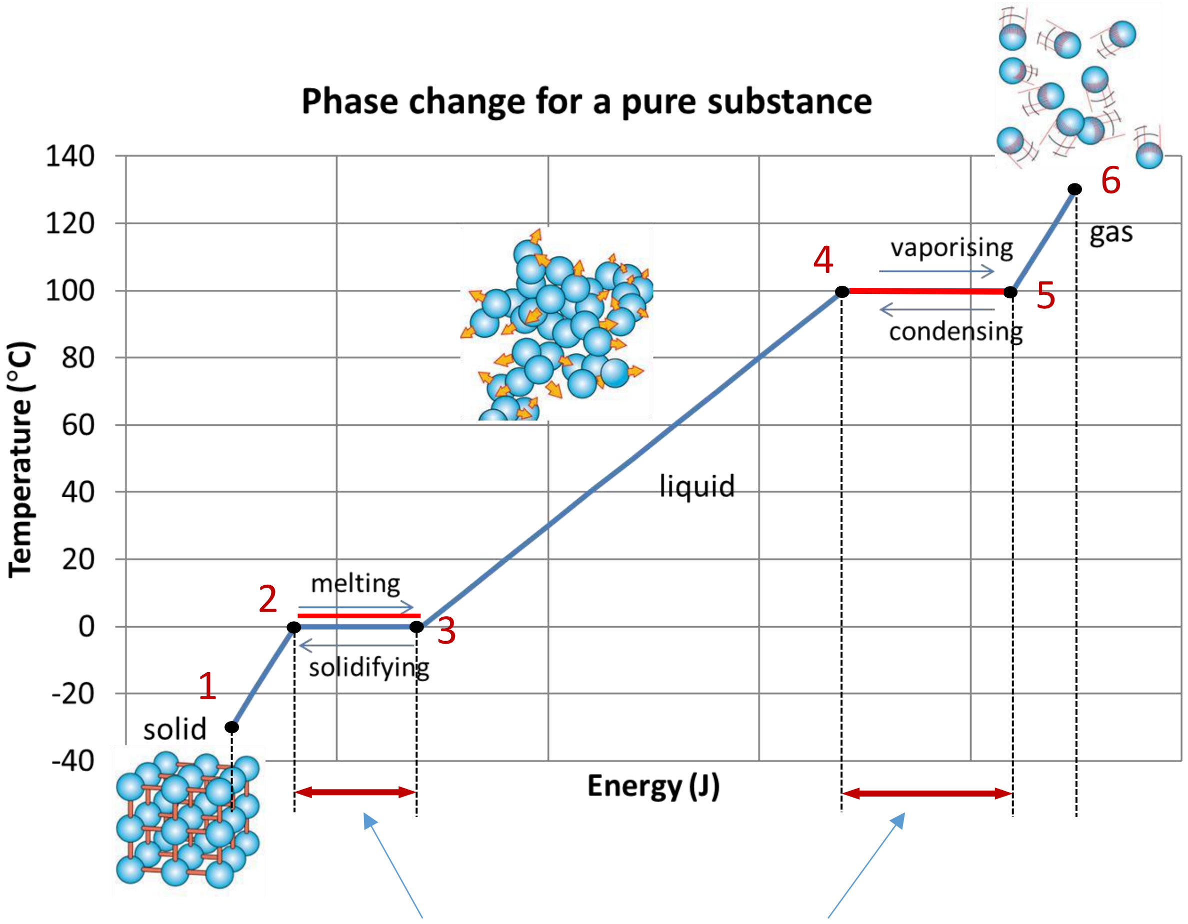
- Consider a pure substance initially as a solid (1).
- As **energy is added** to the solid the **internal energy of the substance increases.**
- The **sensible energy** component of the internal energy increases (temperature increases), as
  - the solid heats up (1→2)
  - the liquid heats up (3→4)
  - the gas heats up (5→6)



change in **sensible** internal energy
(temperature changes

Phase-Change Processes of Pure Substances

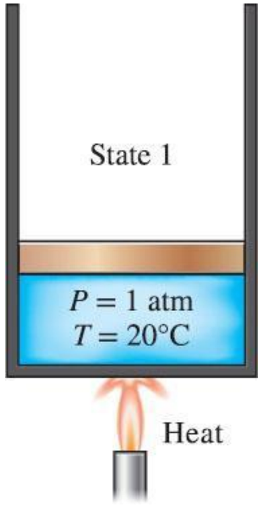
- Consider a pure substance initially as a solid (1).
- As **energy is added** to the solid the **internal energy of the substance increases.**
- The **latent energy** component of the internal energy that increases (change of phase), as:
  - the solid melts (2→3)
  - the liquid vaporises (4→5)



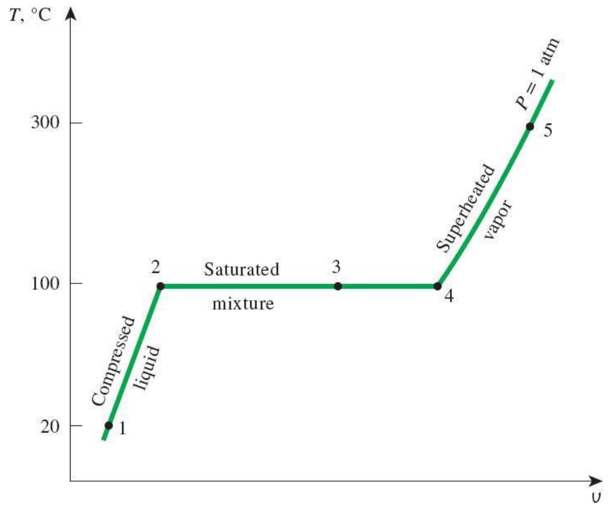
change in **latent** internal energy
(phase changes e.g. solid to liquid)

Phase-Change Processes of Pure Substances

- Mainly focus now on **liquid to gas changes**.
- Consider a **piston cylinder device** containing liquid water at 20°C and 1-atm pressure (= 101.325 kPa).
- The piston maintains a **constant pressure** in the cylinder during the process and **heat energy is applied** to the water.
- In simple terms we find that the water heats up, then begins to boil and ultimately all the liquid water becomes a gas (steam).
- A **plot of the process** on a $T - v$ diagram is shown to the right.
- We will now look at the process in more detail and identify **key terminology** relating to phase change processes



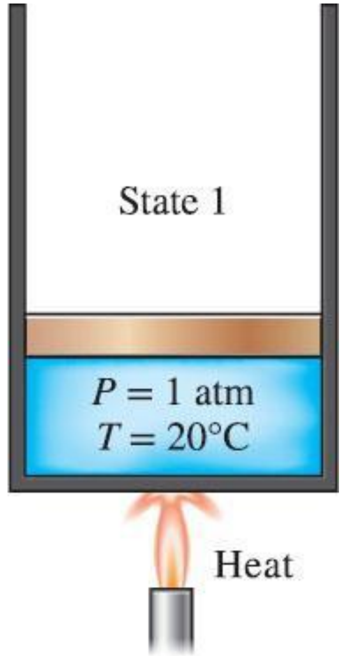
Piston cylinder device

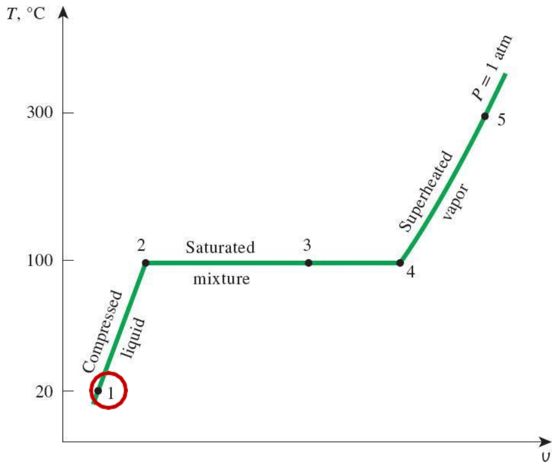


Plot of process

Phase-Change Processes of Pure Substances

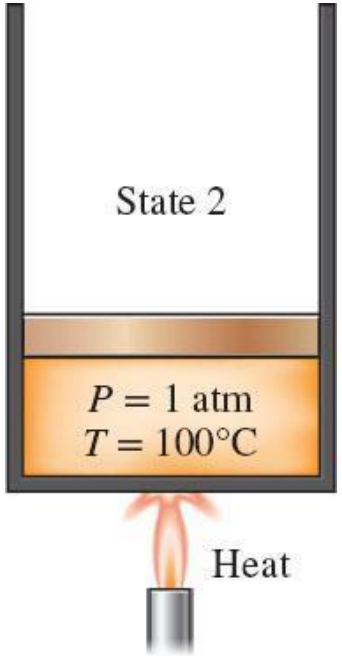
- **State 1**: Water at 1 atm and 20°C exists in the liquid phase and is described as a **compressed liquid**
- A **compressed liquid:** is a substance that is NOT about to vaporise.
- For the given pressure, the temperature < T_{sat}
- The **properties** of a compressed liquid are approximately equal to the properties of a saturated liquid at the same temperature.
- As heat energy is added to the water:
  - the specific volume increases a small amount
  - the temperature rises till it reaches 100°C (state 2).

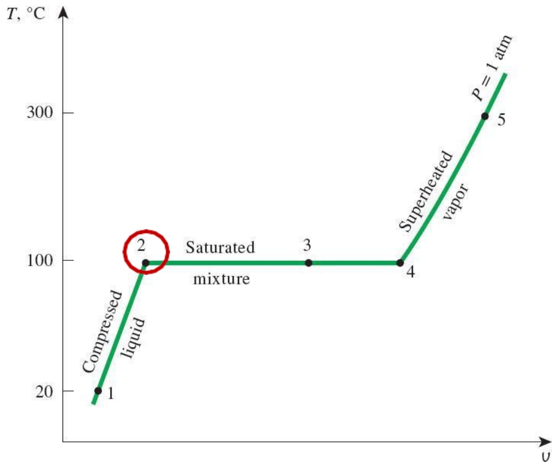




Phase-Change Processes of Pure Substances

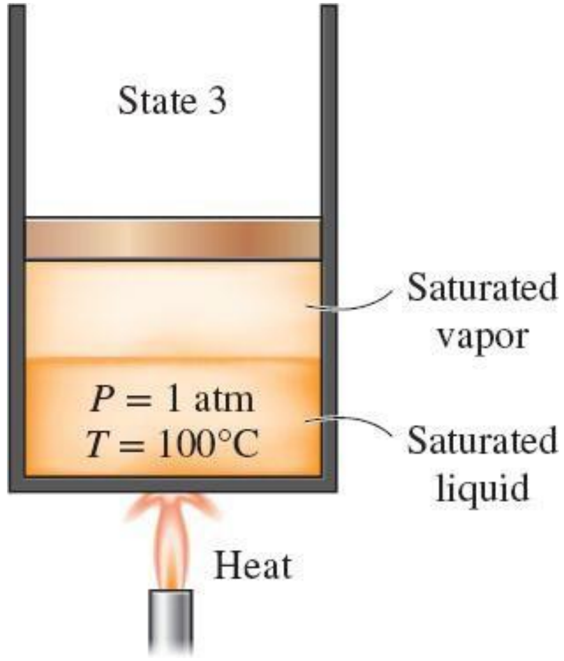
- **State 2**: Water at 1 atm and 100°C exists entirely in the liquid phase and is described as a **saturated liquid**
- **A saturated liquid** is a substance that is about to vaporise (if heat energy is added).
- Temperature = T_{sat}, Pressure = P_{sat}
- Other properties use the subscript f to denote a saturated liquid (= $f$luid) e.g. specific volume of a saturated liquid is v_f, so $v_2 = v_f$.
- As heat energy is added to a saturated liquid some of the liquid will vaporise (boiling begins)
  - Temperature remains constant
  - Specific volume increases significantly





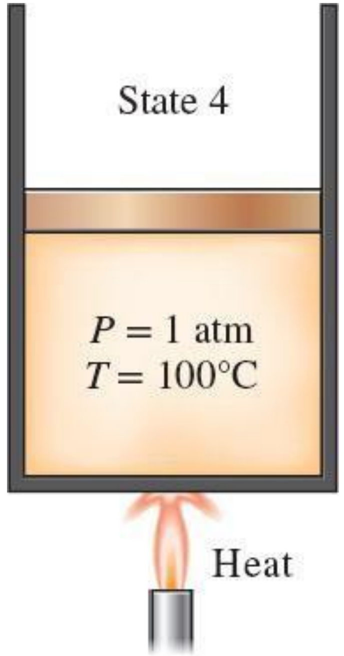
Phase-Change Processes of Pure Substances

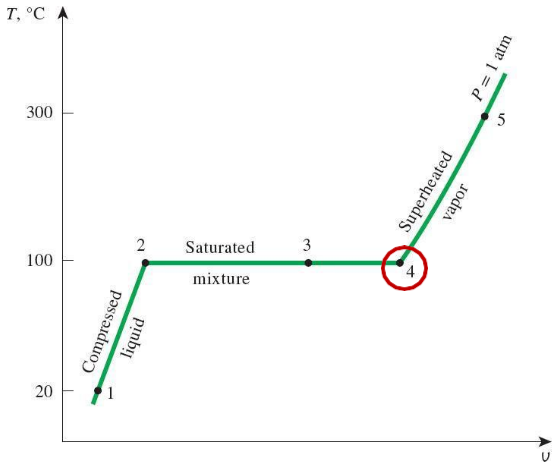
- **State 3**: the water is still at 1 atm and 100°C but a portion is saturated liquid, and a portion is saturated vapour and is called a **saturated liquid-vapour mixture** (or just **saturated mixture**).
- **A saturated liquid-vapour mixture** is where the liquid and vapour (gas) phases coexist in equilibrium.
- Temperature = T_{sat}, Pressure = P_{sat}
- The properties of the saturated mixture depend on the amount vaporised (quality x)
- As heat energy is added more saturated liquid will vaporise (boiling continues).
  - Temperature remains constant
  - Specific volume increases significantly



Phase-Change Processes of Pure Substances

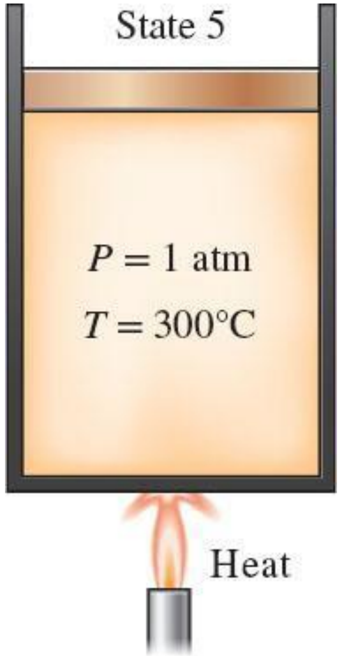
- **State 4**: the water is still at 1 atm and 100°C but now exists entirely in the vapour phase (gas) and is called a **saturated vapour.**
- **A saturated vapour** is a substance that is about to condense (if energy was removed).
- Temperature = T_{sat}, Pressure = P_{sat}
- Other properties use the subscript g to denote a saturated vapour (= $g$as) e.g. specific volume of a saturated vapour is v_g, so $v_4 = v_g$.
- If energy is added to the saturated vapour:
  - Temperature increases
  - Specific volume increases

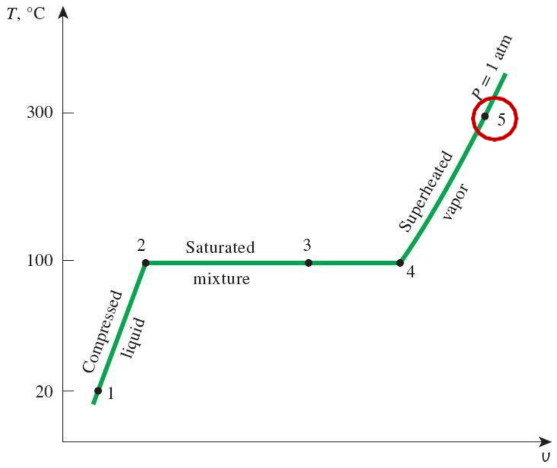




Phase-Change Processes of Pure Substances

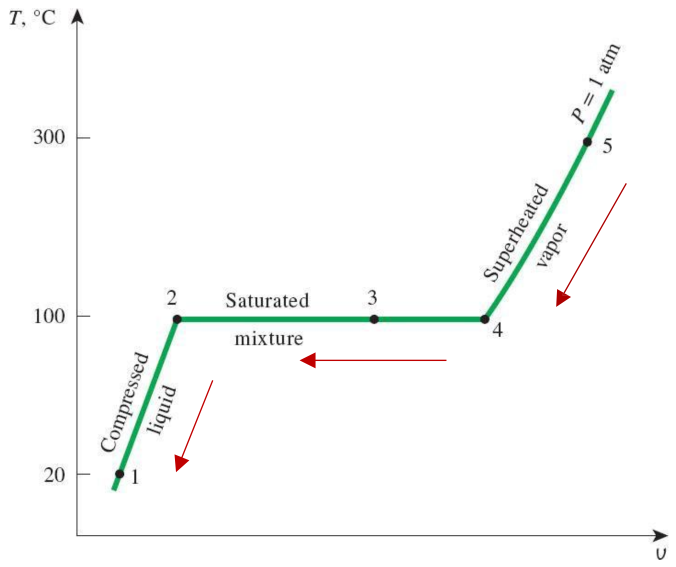
- **State 5**: the water is now at 1 atm and 300°C, exists entirely in the gas phase and is called a **superheated vapour.**
- **A superheated vapour** is a substance that is NOT about to condense (if energy was removed).
- For the given pressure, the temperature > T_{sat}
- Other properties can be found in the superheat tables.
- If energy is added to the superheated vapour:
  - Temperature increases
  - Specific volume increases





Phase-Change Processes of Pure Substances

- If the process just described was reversed i.e. starting at state 5 (water at 1 atm and 300°C) and heat energy *is removed (i.e. the water is cooled)* while maintaining constant pressure, then the water will return to state 1 (1 atm, 20°C) retracing the same path in reverse
- The heat removed during the cooling process will be equal to the heat added during the heating process
- Exercise: what happens when a steel drum that is filled with steam at atmospheric pressure, is sealed and the steam allowed to condense? [Video]



Phase-Change Processes of Pure Substances

Saturation temperature T_{sat} and saturation pressure P_{sat}.

- The temperature at which a liquid boils (changes phase) depends on the surrounding pressure.
- If the surrounding pressure is 1 atm (101.3 kPa) then water will boil at 100°C. At a different pressure the water will boil at a different temperature.
- **Saturation temperature T_{sat}**: the temperature at which a pure substance changes phase at a given pressure.
- **Saturation pressure P_{sat}**: the pressure at which a pure substance changes phase at a given temperature.

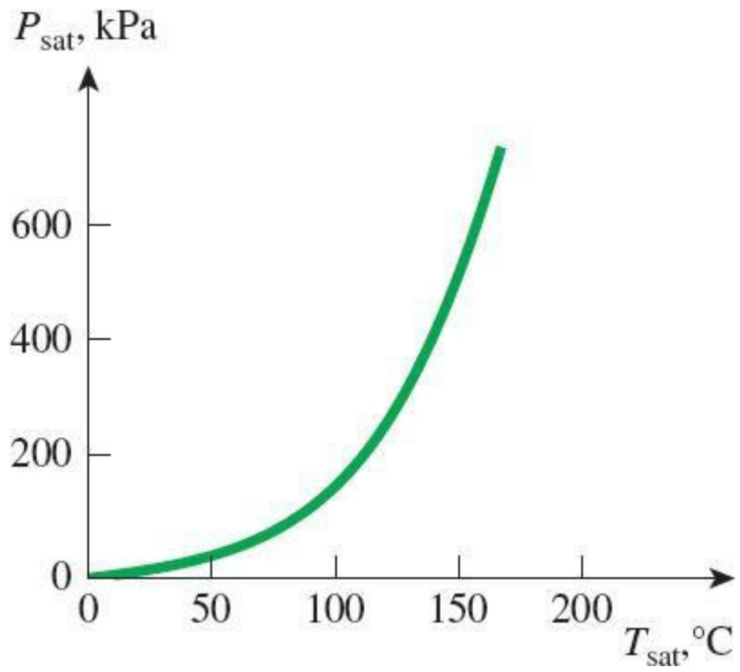


Figure 4–11: The liquid–vapor saturation curve of a pure substance (numerical values are for water).

Phase-Change Processes of Pure Substances

Elevation (m)	Atmospheric pressure (kPa)	Boiling temperature (°C)
0	101.33	100.0
1,000	89.55	96.5
2,000	79.50	93.3
5,000	54.05	83.3
10,000	26.50	66.3
20,000	5.53	34.7

Table 4–2: Variation of the standard atmospheric pressure and the boiling (saturation) temperature of water with altitude.

[Video 1]

[Video 2]

Temperature T (°C)	Saturation pressure P_{sat} (kPa)
0	0.611
5	0.872
10	1.23
15	1.71
20	2.34
25	3.17
30	4.25
40	7.38
50	12.35
100	101.3
150	475.8
200	1554
250	3973
300	8581

Table 4–1: Saturation (or vapor) pressure of water at various temperatures.

Phase-Change Processes of Pure Substances

- It takes a **large amount of energy to melt a solid or vaporise a liquid** i.e. cause it to change phase. For example:
  - To heat 1 kg of water from 0°C to 100°C requires 418 kJ of energy
  - To vaporise (i.e. change phase: liquid to gas) 1 kg of water at 100°C requires 2257 kJ of energy
- The type of internal energy that is absorbed or released during a phase change process is called the latent energy or **latent heat.**
- **Latent heat of fusion:** the amount of energy absorber during melting (or released during solidification (freezing).
- **Latent heat of vaporisation:** the amount of energy absorbed during vaporisation (or released during condensation). Depends on temperature and pressure that phase change occurs
- At 1 atm pressure, for water:
  - Latent heat of fusion (ice ⟷ liquid water) = 334 kJ/kg
  - Latent heat of vaporisation (saturated water ⟷ saturated vapour) = 2257 kJ/kg

Property Diagrams for Phase Change Processes

- The variation of properties during phase-change processes is most easily understood and illustrated on ***property diagrams***.
- Some of the common property diagrams are the $T - v$, $P - v$, and $P - T$ diagrams. (v = specific volume (m^3/kg))
- If we plot the phase change process curve (from above) at different pressures on the $T - v$ diagram, we get Fig. 4-15.
  - The horizontal line (saturated mixture region) gets shorter at higher pressures (and of course higher temperatures)
  - The **critical point** is the state where the saturated vapour and saturated liquid are identical. For water: P_{crit} = 22.06 MPa, T_{crit} = 373.95°C and v_{crit} = 0.003106 m^3/kg
  - At pressures above the critical pressure there is **no distinct phase change process.**

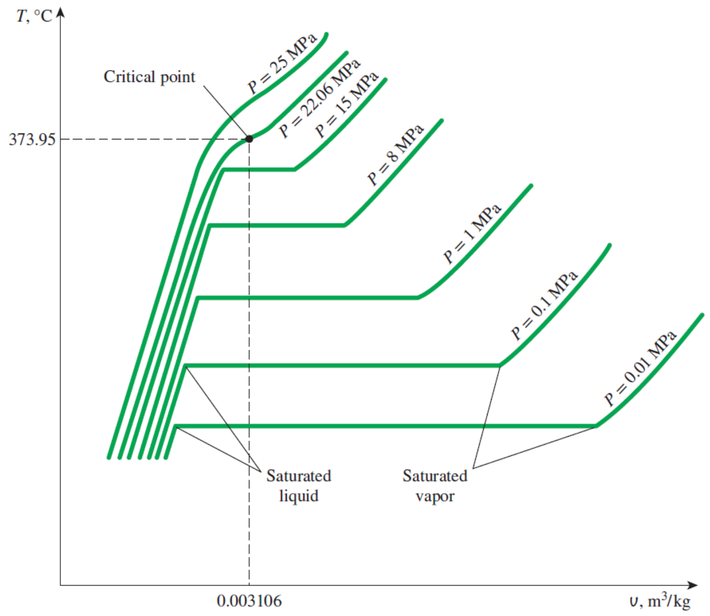


Figure 4–15: *T-v* diagram of constant-pressure phase-change processes of a pure substance at various pressures (numerical values are for water).

Property Diagrams for Phase Change Processes

- The saturated liquid states in Fig. 4-15 can be connected by a line to form the **saturated liquid line** (Fig. 4-17a)
- And the saturated vapour states in Fig. 4-15 can be connected by a line to form the **saturated vapour line line** (Fig. 4-17a)
- These two lines meet at the critical point forming the typical saturation dome shown in Fig. 4-17a

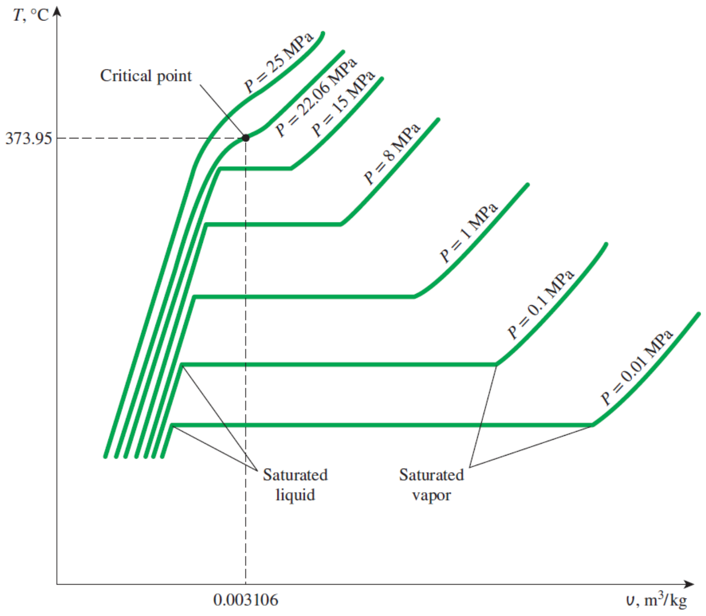


Figure 4–15

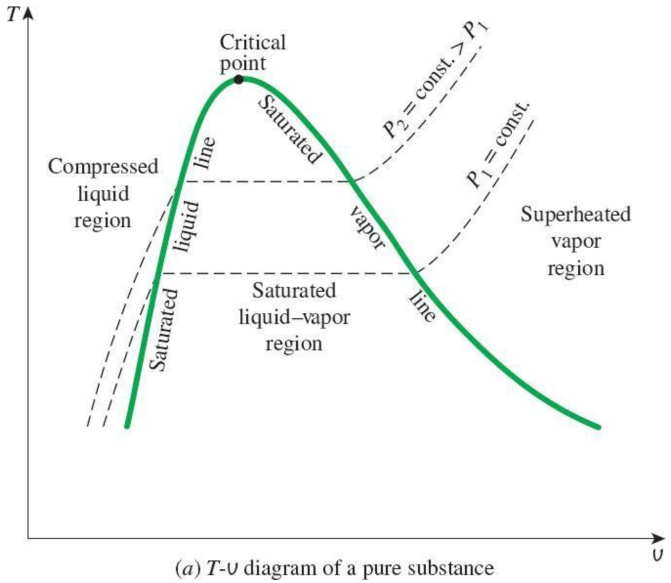


Figure 4–17a

Property Diagrams for Phase Change Processes

The $T - v$ diagram. Note these features:

- Critical point
- Saturation lines (green curves):
  - **Saturated liquid line** – to the left of the critical point.
  - **Saturated vapour line** – to the right of the critical point.
- Regions:
  - **Saturated liquid-vapour region** (or **saturated mixture region**) – the area under the curve between the saturated liquid and saturated vapour lines.
  - **Compressed liquid region** – to the left of the saturated liquid line (also called the **subcooled liquid region**).
  - **Superheated vapour region** – to the right of the saturated vapour.
- Constant pressure lines (isobars)

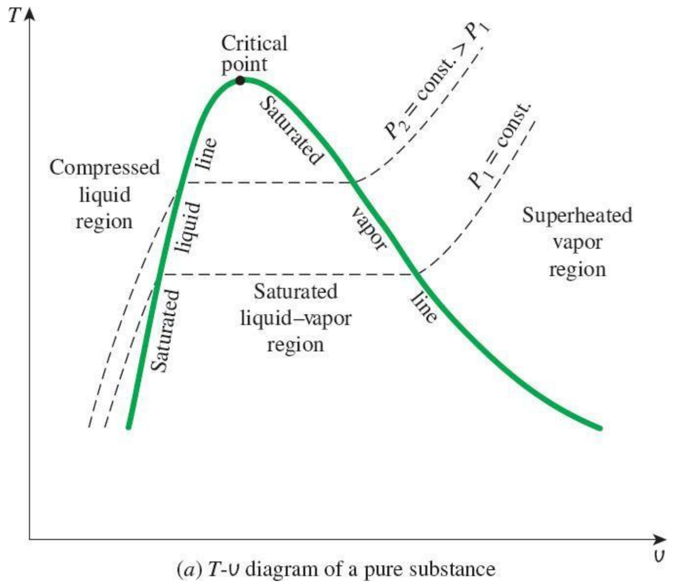


Figure 4–17a: The $T - v$ property diagram of a pure substance.

Property Diagrams for Phase Change Processes

The $P - v$ diagram

- Same general shape as the $T - v$ diagram. Fig. 4-17.
- Same lines and regions as $T - v$ diagram:
  - Critical point
  - Saturated liquid line
  - Saturated vapour line
  - Compressed liquid region
  - Saturated mixture region
  - Superheated vapour region
- Note constant temperature lines (isotherms) have a downward trend.

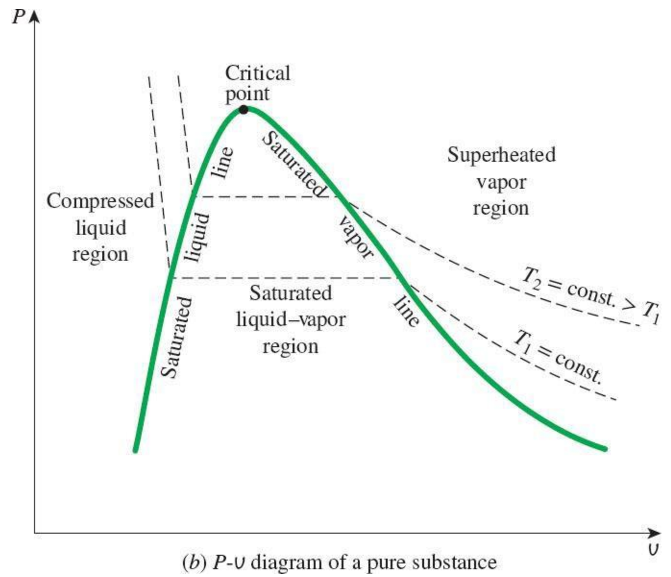


Figure 4–17b: The $P - v$ property diagram of a pure substance.

Property Diagrams for Phase Change Processes

The $P - v$ diagram

- **Constant temperature boiling process on $P - v$ diagram** (Fig. 4-18)
  - Weights on piston on liquid water so that pressure in cylinder is 1 MPa and temperature is constant at 150°C. (compressed liquid)
  - Remove weights (pressure drops) while supplying heat so that temperature remains at 150°C.
  - Specific volume increases and pressure drops till P_{sat} is reached (475.8 kPa) (saturated liquid) then water begins to boil (change phase).
  - Heat added, temp and pressure remain constant at saturated values (saturated mixture) until all the liquid has changed to vapour (saturated vapour).
  - Remove more weights and pressure of vapour further decreases at constant temperature (superheated vapour).

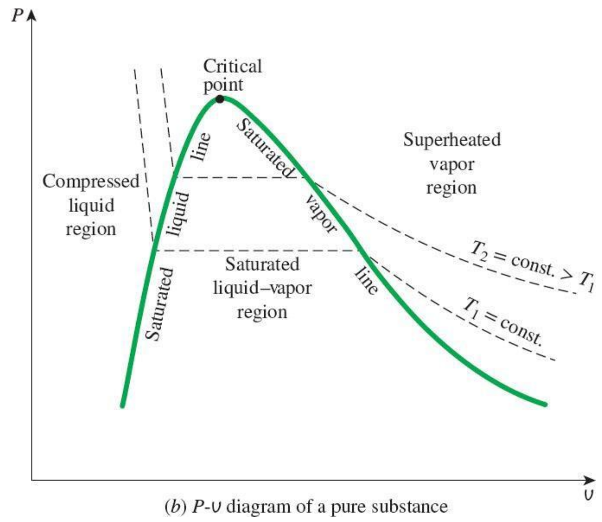


Figure 4–17: The $P - v$ property diagram of a pure substance.

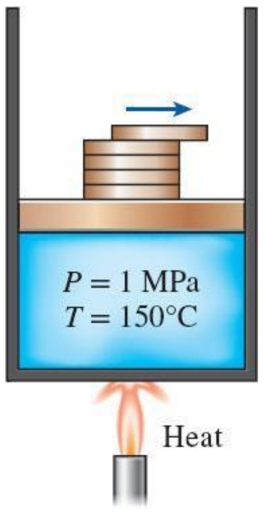
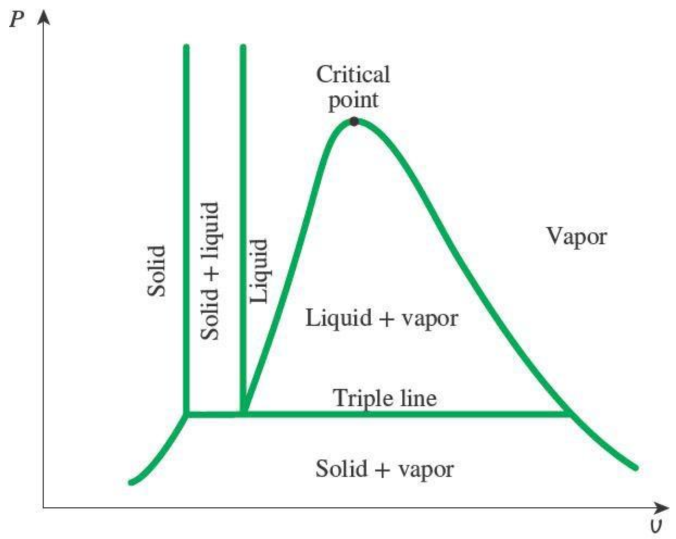


Figure 4–18: The pressure in a piston–cylinder device can be reduced by reducing the weight of the piston.

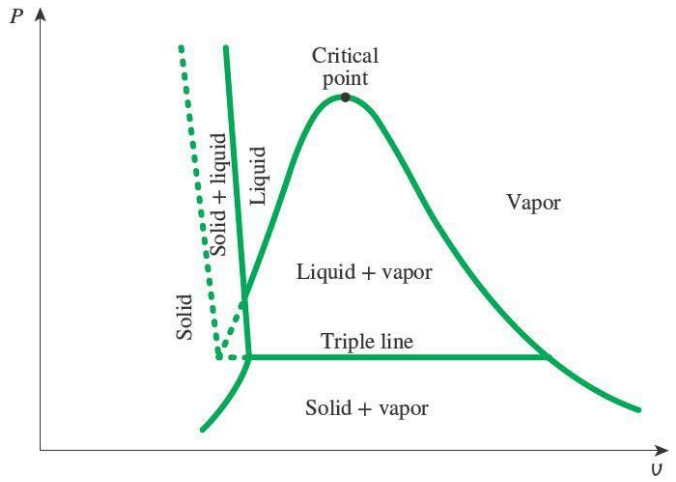
Property Diagrams for Phase Change Processes

Extending the diagrams to include the solid phase

- The $P - v$ and $T - v$ diagrams can be extended to include the **solid phase**.
- The basic principles of phase change, discussed above, also apply to solid – liquid and solid-vapour phase changes.
- Most substances contract on freezing (solidification) - Fig (a), but some, like ice-water, expand - Fig (b).
- Features
  - **Triple line or triple point**. The three principal phases (solid, liquid and gas) exist in equilibrium.
  - **Sublimation**: a heating process where the substance changes phase from a solid to a vapour directly. i.e. the solid evaporates.



(a) P-υ diagram of a substance that contracts on freezing



(b) P-υ diagram of a substance that expands on freezing (such as water)

Property Diagrams for Phase Change Processes

Triple point

- Where all three principle phases (solid, liquid and gas) exist in equilibrium.
- For water: P_{tp} =0.6117 kPa, T_{tp}= 0.01°C

Sublimation

- At low pressures (below triple point) solids evaporate without melting, called sublimation.
- For substances that have a triple point pressure above atmospheric pressure (e.g. dry ice or solid CO_2) they will sublime (change from solid to vapour) at ambient conditions.

Property Diagrams for Phase Change Processes

The $P-T$ diagram

- Sometimes called the **phase diagram** for a substance

Features:

- Sublimation line
- Melting line
- Vaporisation line
- Regions: solid, liquid and vapour
- Triple point and Critical point also shown.

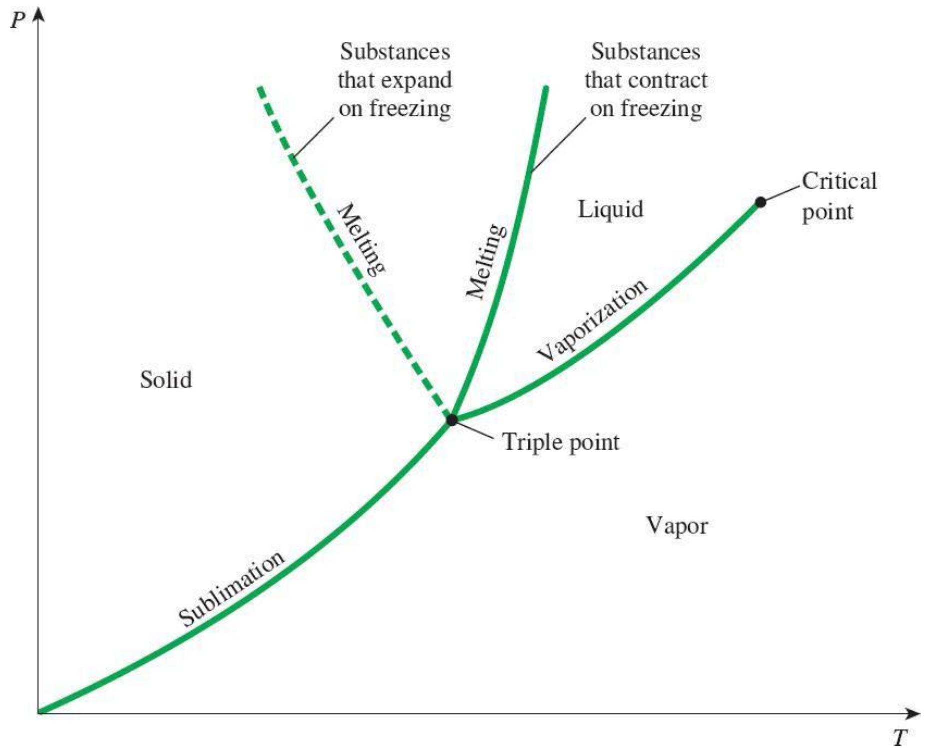


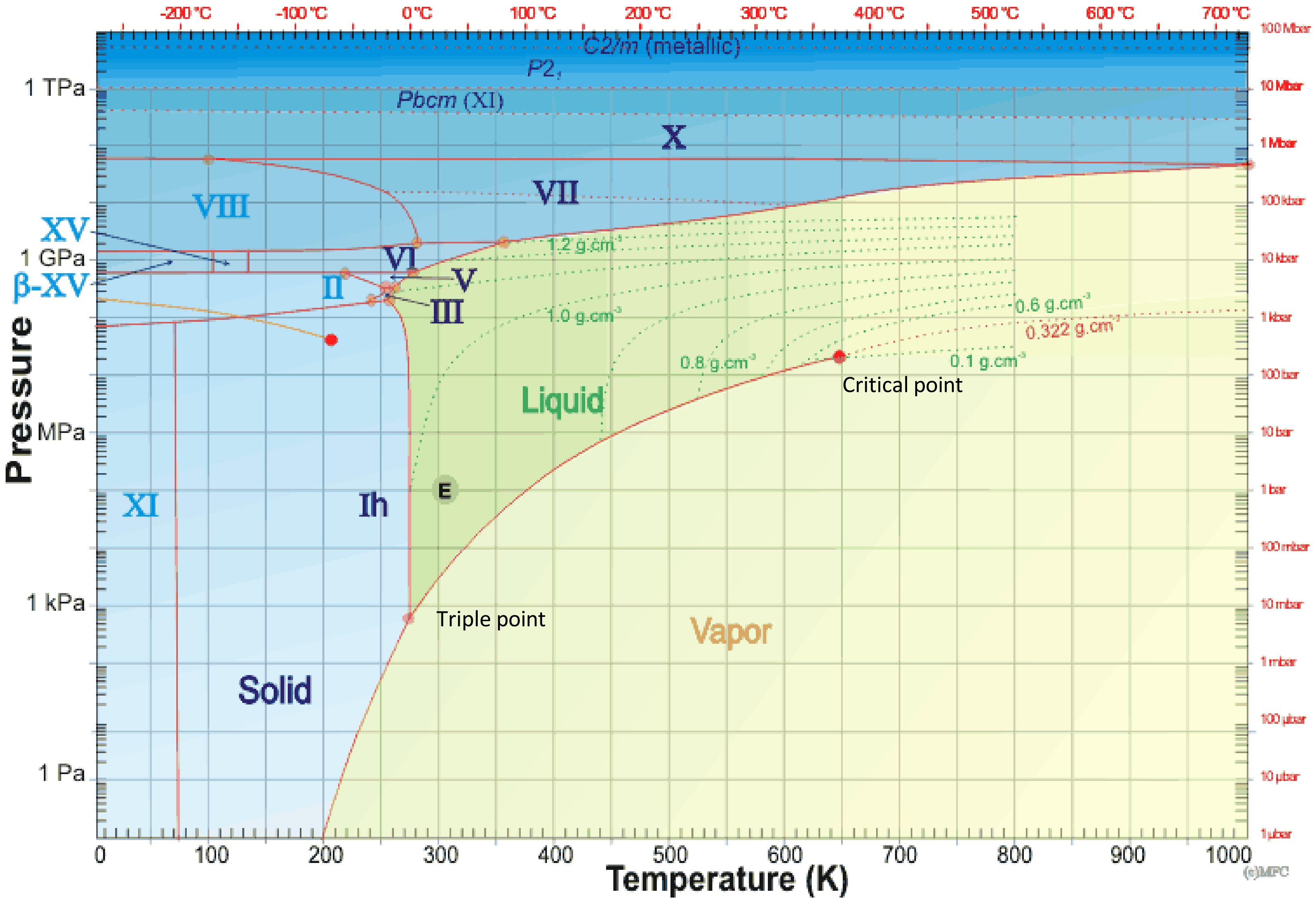
Figure 4–22: *P-T* diagram of pure substances.

Use the $P-T$ diagram to answer these questions:

1. Is it possible to have ice (at equilibrium with its surroundings, i.e. not melting) above the triple point temperature (0.01°C)?
2. Is it possible to have liquid water (at equilibrium) below 0.01°C?
3. Is it possible to have liquid water (at equilibrium) below 0.6117 kPa?
4. Is it possible to have steam (water vapour)(at equilibrium) below 0°C ?

Triple point, water:
P_{tp} =0.6117 kPa, T_{tp}= 0.01°C

Phase diagram – water (for interest only)

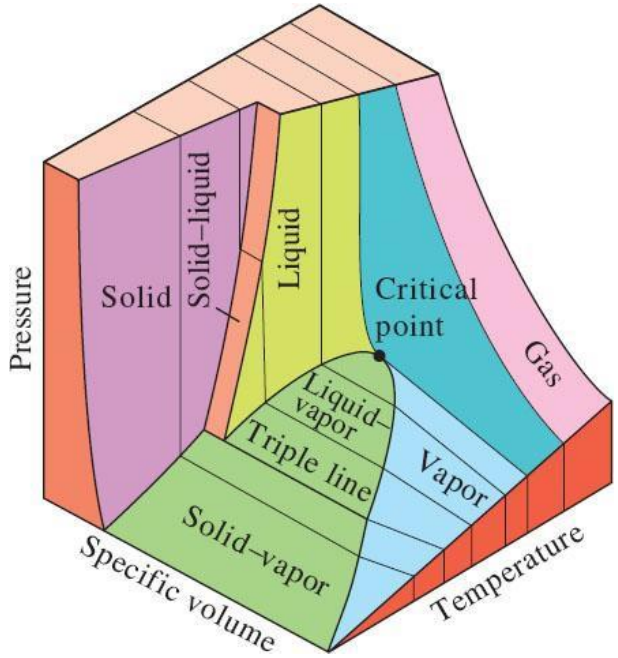


From: [Water phase diagram (lsbu.ac.uk)]

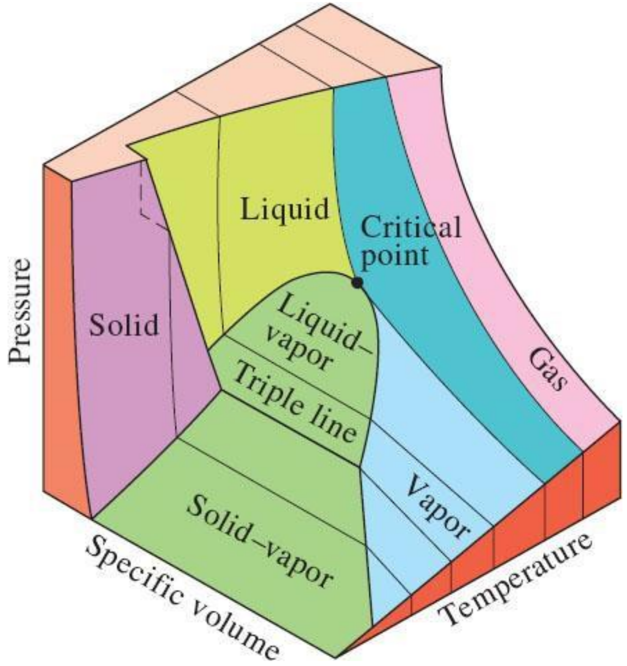
Property Diagrams for Phase Change Processes

The $P - v - T$ surface

- If we plot P, v and T for a pure substance on *x*, *y* and *z* axes we get a surface as shown.
- Only points on the surface are valid equilibrium states



Substances that contract on freezing

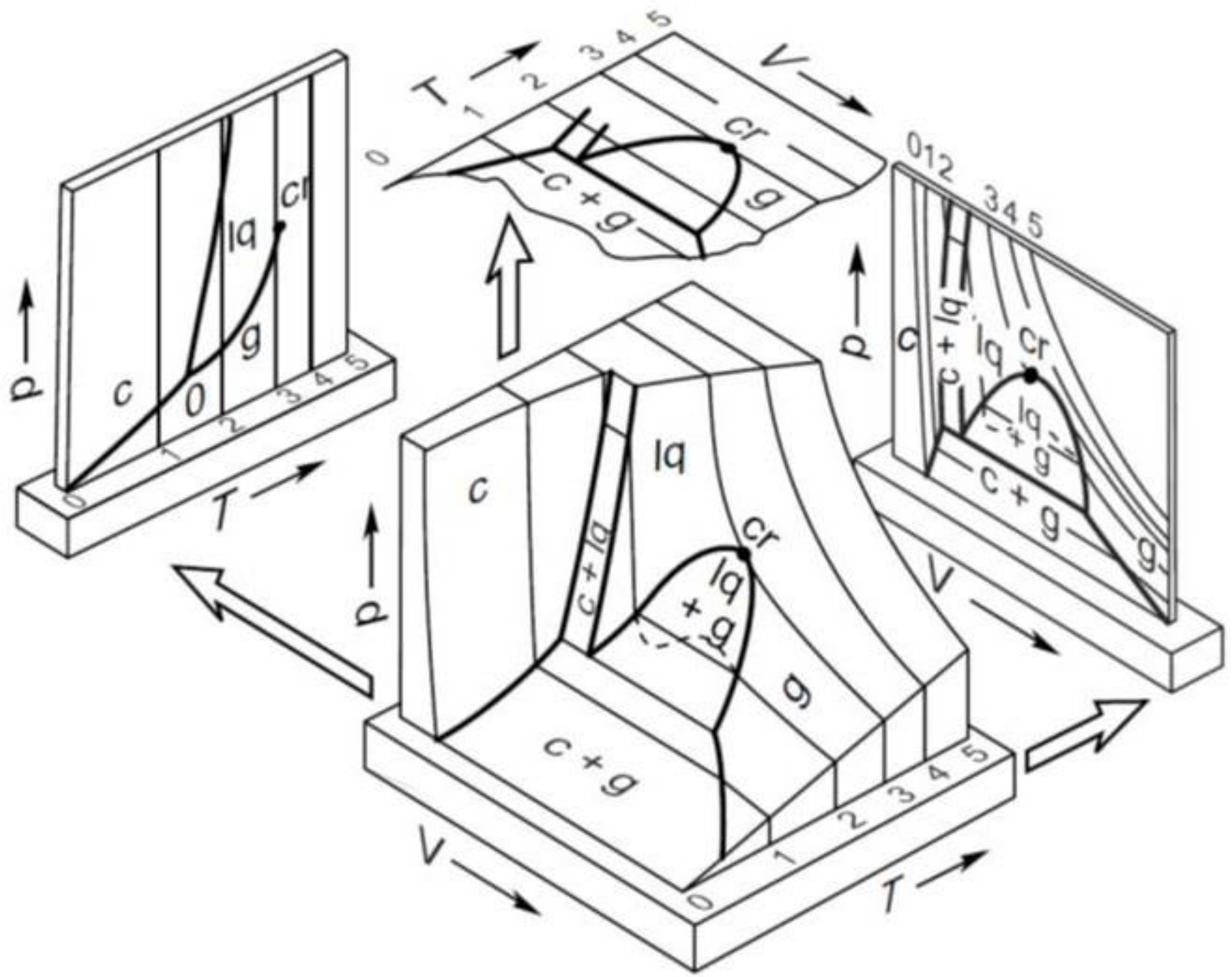


Substances that expand on freezing (like water)

Property Diagrams for Phase Change Processes

The $P - v - T$ surface

- The two-dimensional property diagrams are projections of the $P - v - T$ surface.

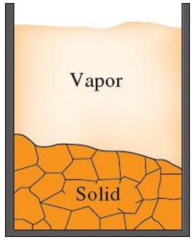


Glasser L, *Water, water everywhere*, Journal of Chemical Education Vol 81 No 3, 2004, 414 - 418

Property Diagrams for Phase Change Processes

Boiling and Evaporation

- Both involve phase change from liquid to vapour… but
- **Evaporation**: occurs at a liquid-vapour interface, no bubble formation or motion.
- **Boiling**: occurs at a solid-liquid interface, characterised by formation and motion of vapour bubbles.
- (Note: sublimation occurs at a solid-vapour interface)

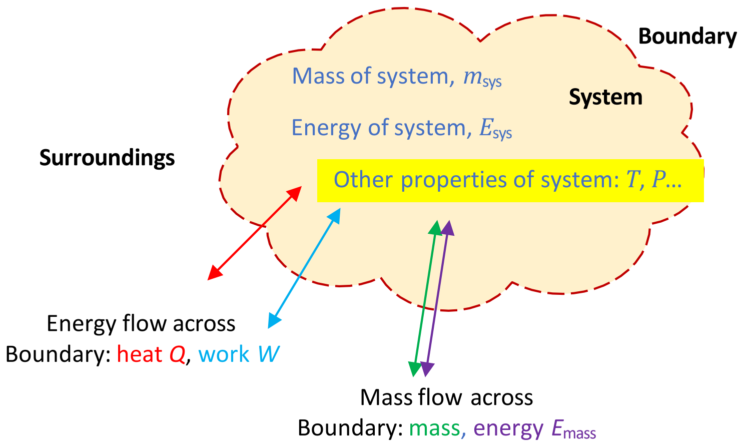


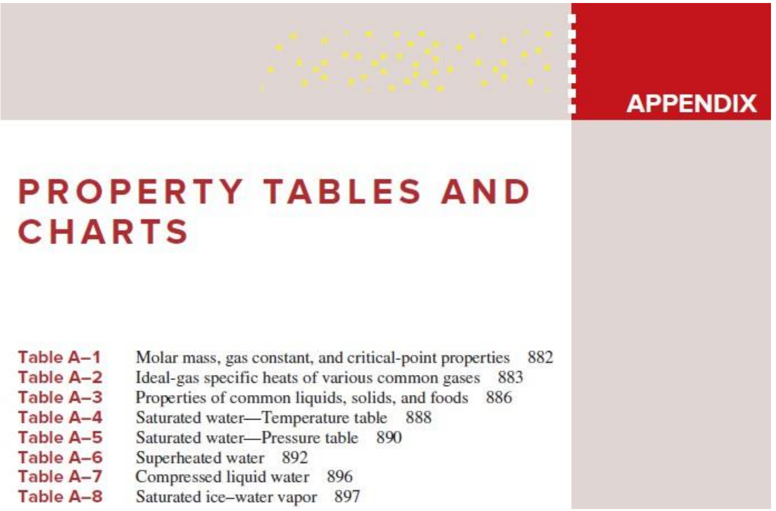
[Credit: Tristan Schmurr | Flickr]

Credit: Roman Sigaev/Fotolia

Property Tables

- During a process the changes of energy in a system also affect the other **thermodynamic properties** of the system (temperature, pressure, specific volume etc).
- *Property diagrams* allow us to *visualise a process* but we also need to be able to **determine the numerical values of the various properties**.
- There are relationships that exist between the properties, but they are often too complex to be expressed by simple equations. (okay for computer)
- Therefore, **thermodynamic properties are often presented in the form of tables**.
- Some properties can be measured easily but others must be calculated using property relations.
- The tables therefore present results of these measurements and calculations.




APPENDIX

PROPERTY TABLES AND CHARTS

Property Tables

- Property Tables can be found in the Appendix of the textbook.

Properties listed in these tables:

- Temperature, T; pressure, P; specific volume, v; internal energy, u.
- Two new properties: enthalpy, h and entropy, s.

PROPERTY TABLES AND CHARTS

TABLE A–4

Saturated water—Temperature table

		Specific volume, m^3/kg		Internal energy, kJ/kg			Enthalpy, kJ/kg			Entropy, kJ/kg·K		
Temp., T °C	Sat. Press., P_{sat} kPa	Sat. liquid, v_f	Sat. vapor, v_g	Sat. liquid, u_f	Evap., u_{fg}	Sat. vapor, u_g	Sat. liquid, h_f	Evap., h_{fg}	Sat. vapor, h_g	Sat. liquid, s_f	Evap., s_{fg}	Sat. vapor, s_g
0.01	0.6117	0.001000	206.00	0.000	2374.9	2374.9	0.001	2500.9	2500.9	0.0000	9.1556	9.1556
5	0.8725	0.001000	147.03	21.019	2360.8	2381.8	21.020	2489.1	2510.1	0.0763	8.9487	9.0249
10	1.2281	0.001000	106.32	42.020	2346.6	2388.7	42.022	2477.2	2519.2	0.1511	8.7488	8.8999
15	1.7057	0.001001	77.885	62.980	2332.5	2395.5	62.982	2465.4	2528.3	0.2245	8.5559	8.7803
20	2.3392	0.001002	57.762	83.913	2318.4	2402.3	83.915	2453.5	2537.4	0.2965	8.3696	8.6661

Property Tables

- **Entropy, s** is a property that relates to the second law of thermodynamics. We will consider it briefly near the end of the Thermodynamics part of this course.
- **Enthalpy, h** is a combination property.
- In the analysis of certain thermodynamic processes, we often encounter **the Pv term** (flow work, flow energy) added to the **internal energy, u**.
  For the sake of simplicity, the combination is defined as a new property, **enthalpy**, and given the symbol h.

$$\boxed{h = u + Pv \qquad (\text{kJTkg})}$$

And

$$\boxed{H = U + PV \qquad (\text{kJ})}$$

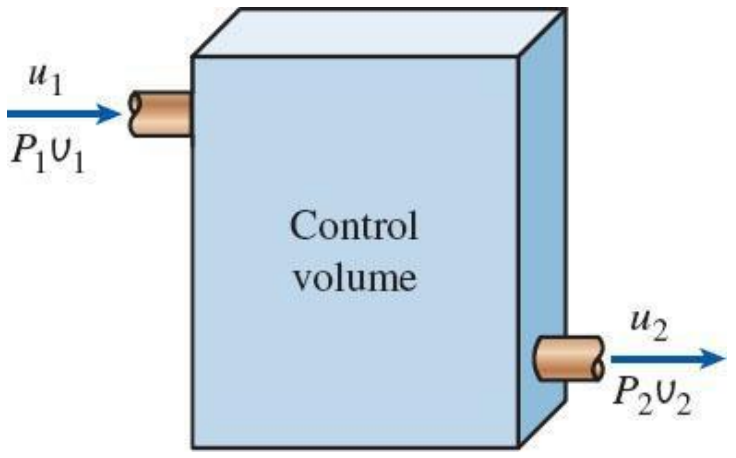
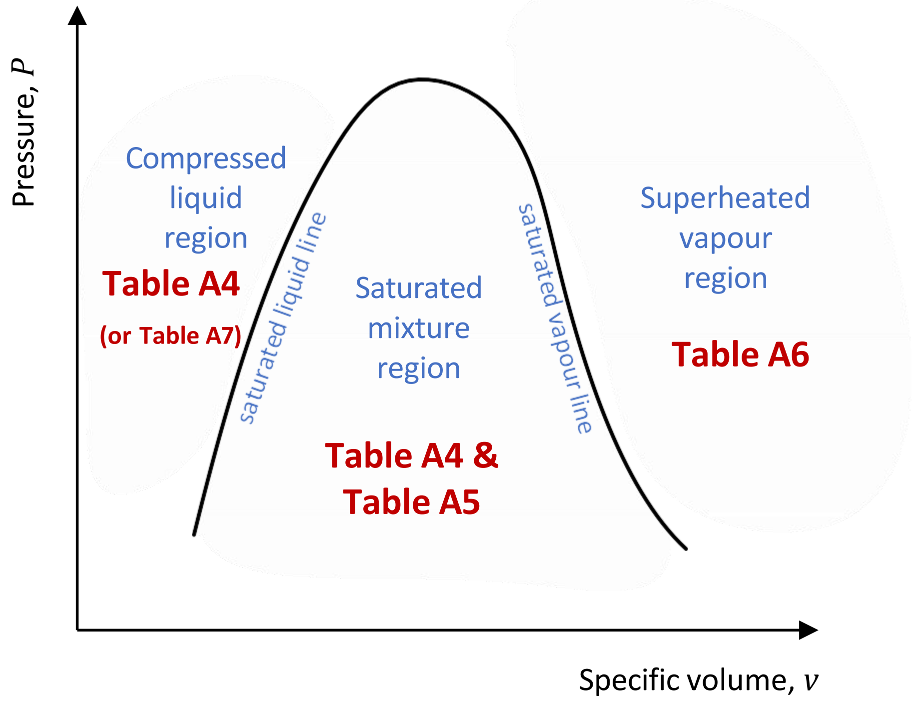


Figure 4–25: The combination $u + Pv$ is often encountered in the analysis of control volumes.

Property Tables

- We will consider the use of **steam tables,** but the principles covered also apply to tables for refrigerants (and any other pure substances)
- There are **different tables for different steam conditions**:
- **Saturated region**: Table A4 and Table A5.
- **Superheated vapour region**: Table A6.
- **Compressed liquid region**: Table A4 (or Table A7 for very high pressures) .
- For **refrigerant R134a** use tables:
  A11 (saturation, T), A12 (saturation, P) and A13 (superheat) in the Appendix.



$P - v$ diagram for **steam** showing saturation lines and relevant property tables

Saturated Liquid-Vapour Tables

- Since temperature and pressure are *dependent* properties in the saturated liquid-vapour region, **two tables are given for the saturation region**.
- **Table A-4** has temperature as the independent property;
- **Table A-5** has pressure as the independent property.
- These two tables **contain the same information** (just arranged differently)

Table A4 – properties in terms of temperature

PROPERTY TABLES AND CHARTS

TABLE A–4

Saturated water—Temperature table

		Specific volume, m^3/kg		Internal energy, kJ/kg			
Temp., T °C	Sat. Press., P_{sat} kPa	Sat. liquid, v_f	Sat. vapor, v_g	Sat. liquid, u_f	Evap., u_{fg}	Sat. vapor, u_g	Sa liq h_f
0.01	0.6117	0.001000	206.00	0.000	2374.9	2374.9	0.(
5	0.8725	0.001000	147.03	21.019	2360.8	2381.8	21
10	1.2281	0.001000	106.32	42.020	2346.6	2388.7	42

Table A5 – properties in terms of pressure

PROPERTY TABLES AND CHARTS

TABLE A–5

Saturated water—Pressure table

		Specific volume, m^3/kg		Internal energy, kJ/kg			
Press., P kPa	Sat. temp., T_{sat} °C	Sat. liquid, v_f	Sat. vapor, v_g	Sat. liquid, u_f	Evap., u_{fg}	Sat. vapor, u_g	Sa lic h_f
1.0	6.97	0.001000	129.19	29.302	2355.2	2384.5	29
1.5	13.02	0.001001	87.964	54.686	2338.1	2392.8	54
2.0	17.50	0.001001	66.990	73.431	2325.5	2398.9	73

So, if you are given the **temperature use Table A4**, and if given the **pressure use Table A5**

Saturated Liquid-Vapour Tables

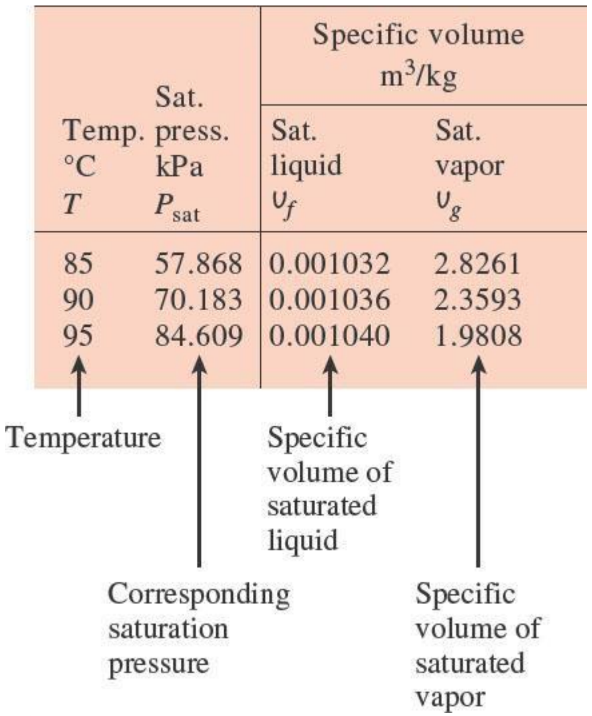
Saturated Liquid and saturated vapour states

- Subscript $\boldsymbol{f}$ denotes properties of the **saturated liquid** e.g. v_f = specific volume of saturated liquid.
- Subscript $\boldsymbol{g}$ denotes properties of the **saturated vapour** e.g. v_g = specific volume of saturated vapour.
- Subscript $\boldsymbol{fg}$ is the **difference** between the saturated liquid and saturated vapour values for that property.

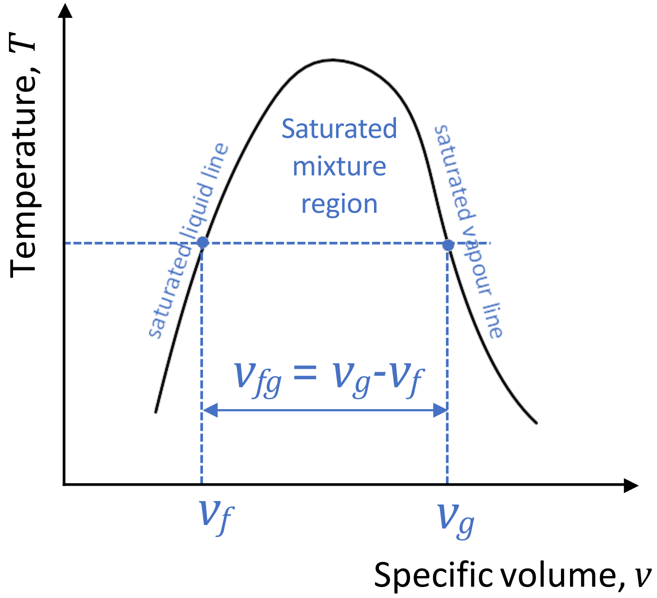
$$\boxed{\begin{aligned} u_{fg} &= u_g - u_f \\ h_{fg} &= h_g - h_f \end{aligned}}$$

- The quantity $\boldsymbol{h_{fg}}$ is called ***the enthalpy of vaporisation*** (latent heat of vaporisation).
  - It represents the energy required to completely vaporise a unit mass of saturated liquid (at a given T or P).
  - It decreases with increasing temperature (or pressure).

Portion of Table A4



Temp. °C T	Sat. press. kPa P_{sat}	Specific volume m^3/kg Sat. liquid v_f	Specific volume m^3/kg Sat. vapor v_g
85	57.868	0.001032	2.8261
90	70.183	0.001036	2.3593
95	84.609	0.001040	1.9808



Saturated Liquid-Vapour Tables

Saturated Liquid-Vapour Mixtures

- During a vaporisation process the substance exists **partly as saturated liquid and partly as saturated vapour**.
- T and P are **dependent properties** in the saturated mixture region so need a new property: quality, x.
- **Quality, x** is the ratio of the mass of saturated vapour to total mass of saturated mixture

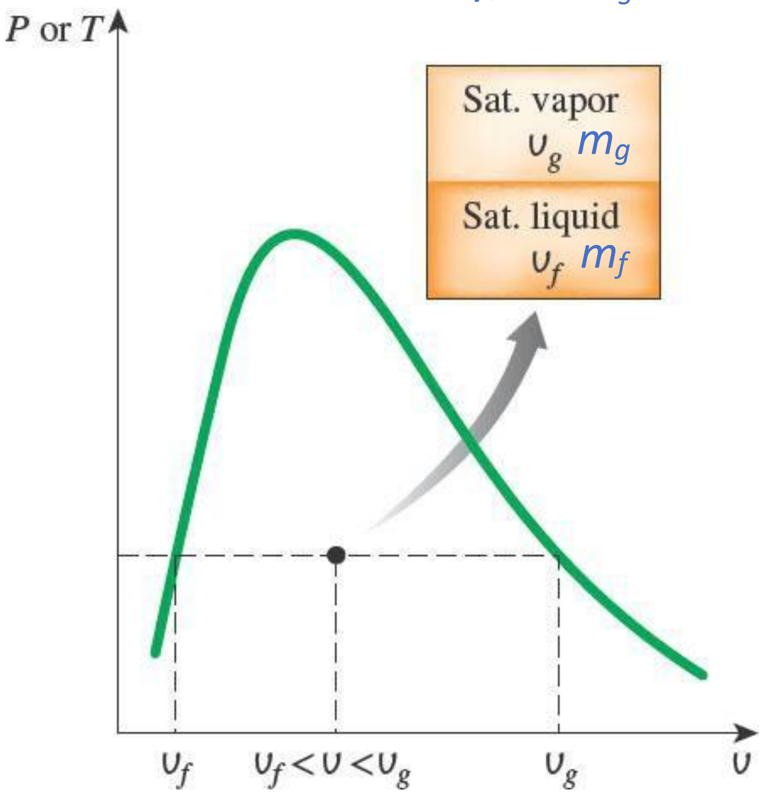
$$x = \frac{m_{\text{sat vapour}}}{m_{\text{total}}} = \frac{m_g}{m_{\text{total}}} = \frac{m_g}{m_f + m_g}$$

- The **quality, x** is sometimes called the dryness fraction and **only applies in the saturation region.**
- x varies from 0 (sat. liquid) to 1 (sat. vapour) or 0% to 100%.

Saturated mixture,

$m_{total} = m_f + m_g$

Quality, $x = m_g / m_{total}$



Saturated Liquid-Vapour Tables

Saturated Liquid-Vapour Mixtures

- Properties (v, u, h, s) in the saturated mixture region can be **determined using the quality, x**.
- Let y be one of these properties at quality x, then the value of y_x can be determined from:

$$\boxed{y_x = y_f + x y_{fg}}$$

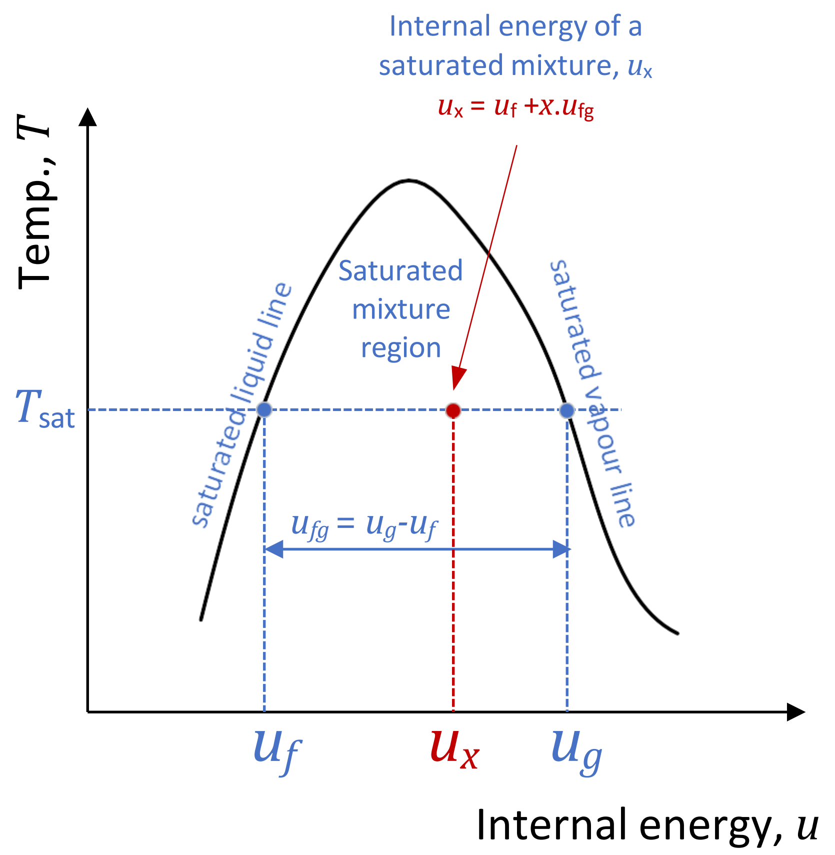
Note the value of y_x will be between the saturated liquid and saturated vapour values, i.e. $y_f \leq y_x \leq y_g$

- So, for internal energy for example:

$$\boxed{u_x = u_f + x u_{fg}}$$

- And if y_x is known the quality, x can be determined:

$$\boxed{x = \frac{y_x - y_f}{y_{fg}}}$$



Superheated Vapour Tables

- In the **superheated vapour region,** the water (steam) is a gas and temperature and pressure are **independent properties.**
- **Table A6** – gives property values in the superheat region.
- Sub-tables show a range of temperatures for a particular pressure.
- Temperature in brackets next to pressure is the saturation temperature.
- First row shows saturation values for that pressure.
- For values between entries in the tables, use linear interpolation.

T, °C	v m^3/kg	u kJ/kg	h kJ/kg
	$P = 0.1$ MPa (99.61°C)		
Sat.	1.6941	2505.6	2675.0
100	1.6959	2506.2	2675.8
150	1.9367	2582.9	2776.6
⋮	⋮	⋮	⋮
1300	7.2605	4687.2	5413.3
	$P = 0.5$ MPa (151.83°C)		
Sat.	0.37483	2560.7	2748.1
200	0.42503	2643.3	2855.8
250	0.47443	2723.8	2961.0

Superheated Vapour Tables

Superheated vapour is characterised by

- Lower pressures ($P < P_{\text{sat}}$ at a given T)
- Higher temperatures ($T > T_{\text{sat}}$ at a given P)
- Higher specific volumes ($v > v_g$ at a given P or T)
- Higher internal energies ($u > u_g$ at a given P or T)
- Higher enthalpies ($h > h_g$ at a given P or T)

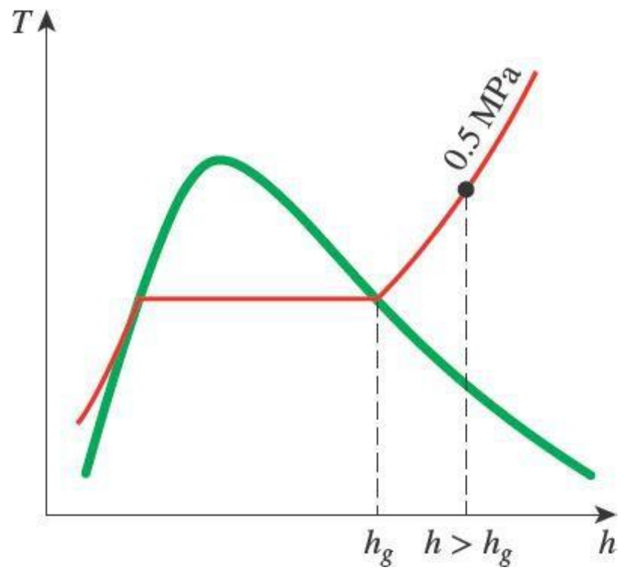


Figure 4–39: At a specified *P*, superheated vapour exists at a higher *h* than the saturated vapour

Compressed Liquid

- When the pressure is greater than P_{sat} for a substance at some temperature T then it is in the **compressed liquid condition**.
- At very high pressures (not common) Table A7 can be used.
- At lower pressures the properties of the compressed liquid are relatively independent of the pressure but **strongly dependent on the temperature**.
- As an approximation we can **treat a compressed liquid like a saturated liquid at the same temperature.**
- So, for any property y (= v, u, h, s) for a compressed liquid at some pressure P and temperature T, its value can be taken as the saturated liquid value of that property at temperature T:

$$\boxed{y \approx y_{f@T}}$$

- For enthalpy h a more accurate approximation can be used. (We will use the above approximation).

$$h \approx h_{f@\,T} + v_f(P - P_{\text{sat}})$$

Compressed Liquid

Compressed liquids are characterised by:

- Higher pressures ($P > P_{sat}$ at a given T)
- Lower temperatures ($T < T_{sat}$ at a given P)
- Lower specific volumes ($v \leq v_f$ at a given P and T)
- Lower internal energies ($u \leq u_f$ at a given P and T)
- Lower enthalpies ($h \leq h_f$ at a given P and T)

For a compressed liquid at a given temperature, T and pressure, P :

$v = v_f$ at temperature T (table A-4)
$u = u_f$ at temperature T (table A-4)
$h = h_f$ at temperature T (table A-4)

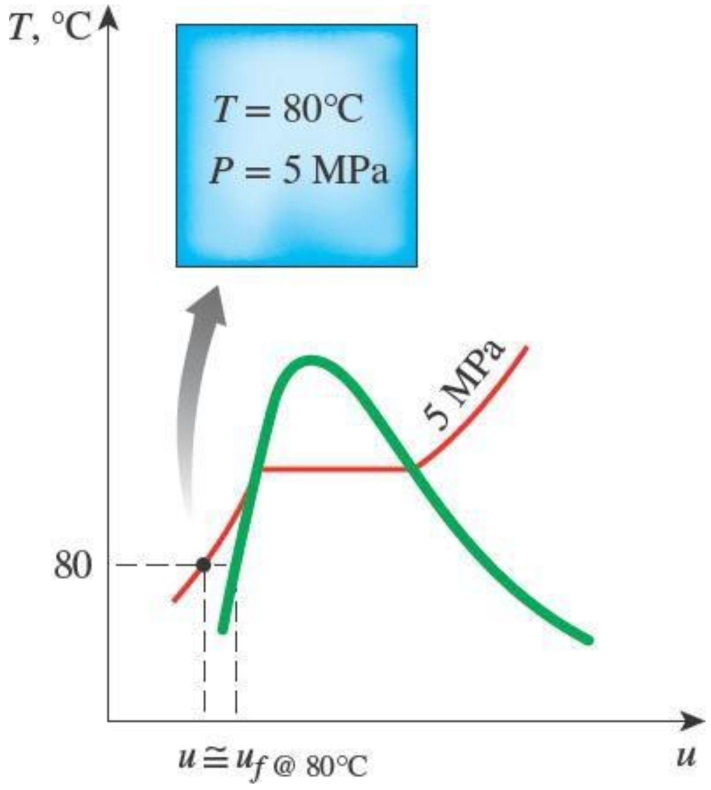


Figure 4–41: A compressed liquid state on a $T - u$ diagram

Reference State

- Property values like u, h and s cannot be measured absolutely.
- So, property values **relative to a reference state** are used
  - And changes in values are calculated from measurable properties and relations between properties.
- The reference state for the water/steam tables is the **triple point of water (0.01°C)** and for **R-134a is -40°C.**
- In thermodynamics we are interested in the **change in properties,** so it doesn't matter what the reference state is.

		Specific volume, m³/kg		*Internal energy,* kJ/kg			*Enthalpy,* kJ/kg			*Entropy,* kJ/kg·K		
Temp., T °C	Sat. Press., P_{sat} kPa	Sat. liquid, v_f	Sat. vapor, v_g	Sat. liquid, u_f	Evap., u_{fg}	Sat. vapor, u_g	Sat. liquid, h_f	Evap., h_{fg}	Sat. vapor, h_g	Sat. liquid, s_f	Evap., s_{fg}	Sat. vapor, s_g
0.01	0.6117	0.001000	206.00	0.000	2374.9	2374.9	0.001	2500.9	2500.9	0.0000	9.1556	9.1556
5	0.8725	0.001000	147.03	21.019	2360.8	2381.8	21.020	2489.1	2510.1	0.0763	8.9487	9.0249

From Table A4: Reference state for water is 0.01°C

Which Table?

- For steam/water or a refrigerant (R-134a), Tables will need to be used to determine any required properties.
- Two independent, intensive properties at least must be given or determined, then using these properties:
    1. Determine the condition of the steam at that state (is it a compressed liquid, a saturated liquid/mixture/vapour or a superheated vapour) – Use *Steam Condition Finder*
    2. Select the correct Table and determine any unknown property values.
- You need to practice determining the steam condition and determining its other properties!